

Education**Institute of Advanced Research**

Ph.D. In Biotechnology (Allied Areas)

Gandhinagar, IN
2020-2025**Nirma University**

M.Sc. in Biochemistry

Ahmedabad, IN
2018-2020**St. Xavier's College**

B.Sc. in Biochemistry

Ahmedabad, IN
2015-2018**Research Experience****Ph.D. Researcher | Institute of Advanced Research****Project Title: Investigate The Interplay Between MDC1 And Beclin-1 In Oxidative Stress And Drug Induced DNA Damage.**

- Project 1 Discovery & Validation of the MDC1-Beclin-1 Axis: Predicted a novel mechanism of chemoresistance using molecular dynamics simulations, and successfully validated this protein-protein crosstalk in vitro under chemotherapeutic stress via co-IP and Western blotting.
- Project 2 Mechanistic Role in Chemoresistance: Utilized high-content imaging and nuclear fractionation to demonstrate that this interaction drives Beclin-1 nuclear accumulation, hyperactivates the DDR, and promotes cell survival under stress.
- Project 3 Identification of Novel Synthetic Vulnerabilities: Developed a combination-therapy strategy by demonstrating that downregulating the DNA repair protein MDC1 increases sensitivity to the CHK1 inhibitor AZD7762, validated through extensive cell based assays.
- Project 4 Interdisciplinary Sensor Development (Collaboration project): Characterized a novel carbazole-based fluorescent ionic liquid for Fe²⁺ sensing in collaboration with a synthetic chemistry team, resulting in a peer-reviewed publication.

Junior Research Fellow | Gujarat Biotechnology Research Center**Project Title: One Health Poultry Hub funded by UK Research and Innovation (UKRI) through the Global Challenges Research Fund (GCRF)**

- Built a characterized library of bacterial isolates from poultry samples to support downstream antimicrobial resistance (AMR) analysis.
- Identified bacterial species using biochemical assays and 16S rRNA sequencing.

Master's Dissertation | Nirma University**Project: Identification and cloning of novel Rab9 homologue in Entamoeba histolytica**

- Identified a novel Rab9 homologue using bioinformatics pipelines and subsequently validated its expression in engineered Entamoeba trophozoites via molecular cloning and transfection studies

Summer Intern | Dept. of Quality Control at Amneal Pharmaceuticals Pvt. Ltd.

- Performed sterility testing and microbial contamination analysis on raw materials, in-process samples, and finished drug products in a pharmaceutical manufacturing environment.

Publications

1. **Pandya Kavya**, and Neeru Singh. "MDC1 promotes nuclear localization of Beclin-1 and supports its role in ATM pathway in response to oxidative stress." European Journal of Cell Biology (2026): 151540.
2. **Pandya Kavya**, and Neeru Singh. "Deciphering the molecular links between DNA damage response and autophagy in cancer: Implications for chemotherapy resistance." Cancer Research 86, no. 7_Supplement (2026): 1739-1739.
3. **Pandya Kavya**, and Neeru Singh. "Unraveling the molecular interactions between DDR and autophagy in response to oxidative stress." Cancer Research 85, no. 8_Supplement_1 (2025): 4198-4198.
4. Singh Neeru, and **Kavya Pandya**. "Abstract 1021 Decoding the DDR and Autophagy cross talk, a cellular response to genotoxic stress and its implications in cancer therapy." Journal of Biological Chemistry 301, no. 5 (2025).
5. Vyas Saurabh, **Kavya Pandya**, Neeru Singh, and Roli Mishra. "A turn off fluorescent ionic liquid with carbazole fluorophore for sensing Fe²⁺ ions." Journal of Molecular Structure 1342 (2025): 142732.
6. Rathore Sangeeta, Rohi Bhatt, Swati Joshi, **Kavya Pandya**, and Neeru Singh. "Applications and associated challenges of CRISPR-Cas technology in agriculture." In CRISPR-Cas System in Translational Biotechnology, pp. 265-280. Academic Press, 2024.
7. **Pandya Kavya**, and Neeru Singh. "In silico study reveals unconventional interactions between MDC1 of DDR and Beclin-1 of autophagy." Molecular Diversity 27, no. 6 (2023): 2789-2802.
8. **Pandya Kavya**, Deep Jagani, and Neeru Singh. "CRISPR-cas systems: programmable nuclease revolutionizing the molecular diagnosis." Molecular biotechnology 66, no. 8 (2024): 1739-1753.
9. **Kavya Pandya**, Jagani Deep, Panchal Charmi and Singh Neeru. MDC1 Downregulation Enhances the Efficacy of CHK1 Inhibitor AZD7762 in Cervical Cancer Cells (Under Communication)
10. **Kavya Pandya** and Neeru Singh. CRISPR-Cas System: Molecular Scissors Revolutionizing Genetic Engineering. Natural Sciences (2022) (Institutional Journal of Institute of Advanced Research)

Honors & Awards

• Lady Tata Memorial Trust Research Fellowship	2022-2025
• DBT Junior Research Fellowship, awarded by Government of India	2021
• GATE (Life Sciences), All India Rank 710	2020
• Gujarat State Eligibility Test	2020
• SHODH Fellowship, awarded by Government of Gujarat	2021

Technical Skills

Molecular & Cell Biology

- **Cell Culture:** Mammalian cell culture (HeLa, H358, H1373, A549, MCF7, SKOV3, UWB BRCA+/-) parasite culture
- **Protein Analysis:** Protein expression and isolation, SDS-PAGE, Western blotting, Co-immunoprecipitation (Co-IP)
- **Cellular Analysis:** Immunocytochemistry, Flow cytometry, Nuclear fractionation, Apoptosis analysis (Annexin V-PI), Clonogenic assay, and Cell cycle (Propidium-Iodide)
- **Molecular Techniques:** PCR, qPCR, molecular cloning, gene knockdown (shRNA)/knockin, Stable/Transient Overexpression, protein overexpression, Overexpression system and transfection
- **Cell Based Assays:** MTT, CTG assay and ADP Glo assay

Bioinformatics & Computational Biology

- **Simulation and Modeling:** Molecular docking, protein-protein docking, ligand docking, Molecular dynamics simulation (Schrodinger and Gromacs), homology modeling
- **Analysis and Visualization Tools:** Chimera and Pymol
- **Sequence Analysis:** Clustal W, BLAST, APE, Chromas, Pfam

Instrumentation

- Flow cytometry, Fluorescence microscope, Image cytometer
- PCR, ELISA reader, luminometer, spectrophotometry, nanodrop
- Gel electrophoresis, Gel Documentation system

Conferences

• Poster presentation titled “Deciphering the molecular links between DNA damage response and autophagy in cancer: Implications for chemotherapy resistance” at AACR Annual Meeting, San Diego	2026
• Poster presentation titled “Unraveling the molecular interactions between DDR and autophagy in response to oxidative stress” at AACR Annual Meeting, Chicago	2025
• Poster presentation titled “Structural insights into the DDR-Autophagy nexus: In silico study uncovering interaction between MDC1 and Beclin-1” at MedBiotech Conference, Gujarat Biotechnology University, Gandhinagar	2024
• Oral presentation titled “Investigate The Novel Role Of MDC1 As An Autophagy Inducer And Its Implications In Tumorigenesis” at Annual Research & Innovation Conclave, Institute of Advanced Research, Gandhinagar- Best Oral Presentation Award	2021

Workshops & Trainings

• Coordinated and led “Animal Cell Culture Techniques and Apoptosis Detection Assays,” funded by GSBTM	2022
• Flow cytometry workshop, Nirma University	2020
• CRISPR-Cas9 workshop, Nirma University	2019
• Learned basics of cytogenetic techniques including FISH, karyotyping, and automated Sanger sequencing at Supratech Laboratories	2019

Peer Review

- Invited Peer Reviewer, Journal of Biomolecular Structure and Dynamics (Taylor & Francis)
- Invited Peer Reviewer, Cell Biochemistry and Biophysics (Springer Nature)

Affiliations

American Association of Cancer research (AACR) | American Society of Cell Biology (ASCB) | American Society of Clinical Oncology (ASCO)

Contact Information for Referees

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